

Backtesting

Proving an Idea Before Risking Real Money — Lesson Explainer

The Concept

A backtest takes a strategy's precisely defined rules and simulates applying them to historical data. It answers 'if I had followed this exact rule in the past, what result would it have produced?' - a necessary first filter before risking real capital.

Two failure modes dominate: lookahead bias (using information that wouldn't have been available at the time) and overfitting (tuning rules so precisely to historical data that the strategy has memorized the past rather than found a real, repeatable pattern).

A realistic backtest also accounts for commissions, the bid-ask spread, and slippage. A strategy that looks great with zero costs can turn unprofitable once realistic frictions are added.

Illustrative Example: Same Backtest, With and Without Realistic Costs

Version	Trades/Yr	Gross Return	Est. Cost Drag	Net Return
No Costs Assumed	240	+18.0%	0.0%	+18.0%
Realistic Costs Assumed	240	+18.0%	-11.5%	+6.5%

240 trades a year at even a small per-trade cost compounds into a large annual drag - the 'no costs' version looks like a great strategy, the realistic version looks merely decent. Always ask which one you're looking at.

5 Things to Know About Backtesting

1. A great backtest is necessary, not sufficient - it filters bad ideas, doesn't guarantee good ones.
2. Overfitting gets worse the more you tune - more adjusted parameters means more risk of fitting noise.
3. Out-of-sample testing is the standard defense - test rules on data they never saw during tuning.
4. Survivorship bias can silently inflate results - testing only on companies that still exist today.
5. A backtest is a simulation, not a guarantee - conditions change, and history never repeats exactly.

4 Things to Check in Any Backtest

- Check for lookahead bias - confirm every input would genuinely have been known at that time.
- Include realistic costs - model commissions, spread, and slippage, especially for frequent trading.
- Hold out a true test period - never tune parameters on the same data used to judge the result.
- Stress-test across regimes - check performance separately across calm, trending, volatile periods.

Worth knowing:

this explains backtesting methodology using illustrative numbers - it isn't personalized financial advice, and no strategy or backtest result here is a recommendation to trade. Past backtested performance is never a guarantee of future results.

Quiz — Answer the questions, then check the key on the next page

1. What does a backtest do?

- A. Predicts exactly what a strategy will return in the future
- B. Simulates a strategy's precisely defined rules against historical data
- C. Automatically executes trades in a live account
- D. Guarantees a strategy is profitable if the result is positive

2. What is lookahead bias?

- A. Trading too infrequently
- B. Using too many years of historical data
- C. Accidentally using information that wouldn't have been available at the time
- D. Ignoring trading costs in a simulation

3. In the illustrative example, why did realistic costs matter so much for the frequent-trading strategy?

- A. Because the strategy only traded once a year
- B. Because costs only apply to losing trades
- C. Because commissions are the same regardless of trade frequency
- D. Because 240 trades a year compounds even a small per-trade cost into a large annual drag

4. What is the standard defense against overfitting described in this lesson?

- A. Out-of-sample testing - testing rules, unchanged, on data they never saw during tuning
- B. Adding more parameters to the strategy
- C. Testing only on the most recent month of data
- D. Ignoring the backtest result entirely

5. What is survivorship bias in this context?

- A. Only trading strategies that have never lost money
- B. Backtesting only during bull markets
- C. Testing only on companies that still exist today, ignoring ones that went bankrupt or were delisted
- D. Using too short a historical data window

Answer Key

1. What does a backtest do?

Correct answer: B. Simulates a strategy's precisely defined rules against historical data

A backtest answers 'if I had followed this exact rule in the past, what would have happened?' - not a guarantee.

2. What is lookahead bias?

Correct answer: C. Accidentally using information that wouldn't have been available at the time

Lookahead bias inflates a backtest's apparent result by using data before it would genuinely have been known.

3. In the illustrative example, why did realistic costs matter so much for the frequent-trading strategy?

Correct answer: D. Because 240 trades a year compounds even a small per-trade cost into a large annual drag

A high trade count multiplies small per-trade frictions into a substantial annual cost.

4. What is the standard defense against overfitting described in this lesson?

Correct answer: A. Out-of-sample testing - testing rules, unchanged, on data they never saw during tuning

Out-of-sample testing checks whether rules still hold up on a separate period they never influenced.

5. What is survivorship bias in this context?

Correct answer: C. Testing only on companies that still exist today, ignoring ones that went bankrupt or were delisted

Survivorship bias inflates results by silently excluding companies that failed along the way.